

Microeconomics II

What do they have in common?



Topic 9: Oligopoly

Matěj Bajgar
matej.bajgar@fsv.cuni.cz

Introduction

- **Oligopoly** = a market with several competitors
 - $N^{\text{MONOPOLY}} < N^{\text{OLIGOPOLY}} < N^{\text{PERFECT COMPETITION}}$
 - Firms affect price
 - Strategic interaction between firms
 - **Duopoly** = oligopoly with 2 firms

Examples of oligopolistic markets?

Introduction

- **Oligopoly** = a market with several competitors
 - $N^{\text{MONOPOLY}} < N^{\text{OLIGOPOLY}} < N^{\text{PERFECT COMPETITION}}$
 - Firms affect price
 - Strategic interaction between firms
 - **Duopoly** = oligopoly with 2 firms
- Many **types of settings**
 - Simultaneous vs. sequential decisions
 - Price vs. quantity setting
 - Homogenous vs. diversified product
 - Competitive vs. cooperative

Roadmap

- Homogenous products
 - Sequential quantity setting (the **Stackelberg** model)
 - Simultaneous quantity setting (the **Cournot** model)
 - Simultaneous price setting (the **Bertrand** model)
 - Cooperative game (**Collusion**)
- Differentiated products
 - Simultaneous price setting (the **Bertrand** model)
 - the **Hotteling** model
 - **Monopolistic competition**

Motivation

- Most markets can be described as oligopolies



amazon

- Foundation for algorithmic pricing, high-frequency pricing...



- The bread and butter of anti-trust



Sequential quantity setting (Stackelberg)

Two firms

- **Quantity leader (firm 1):** a dominant firm who sets the quantity.
- **Quantity follower (firm 2):** takes the quantity of the leader as given.

Examples of quantity leaders

A



B



C



Sequential quantity setting (Stackelberg)

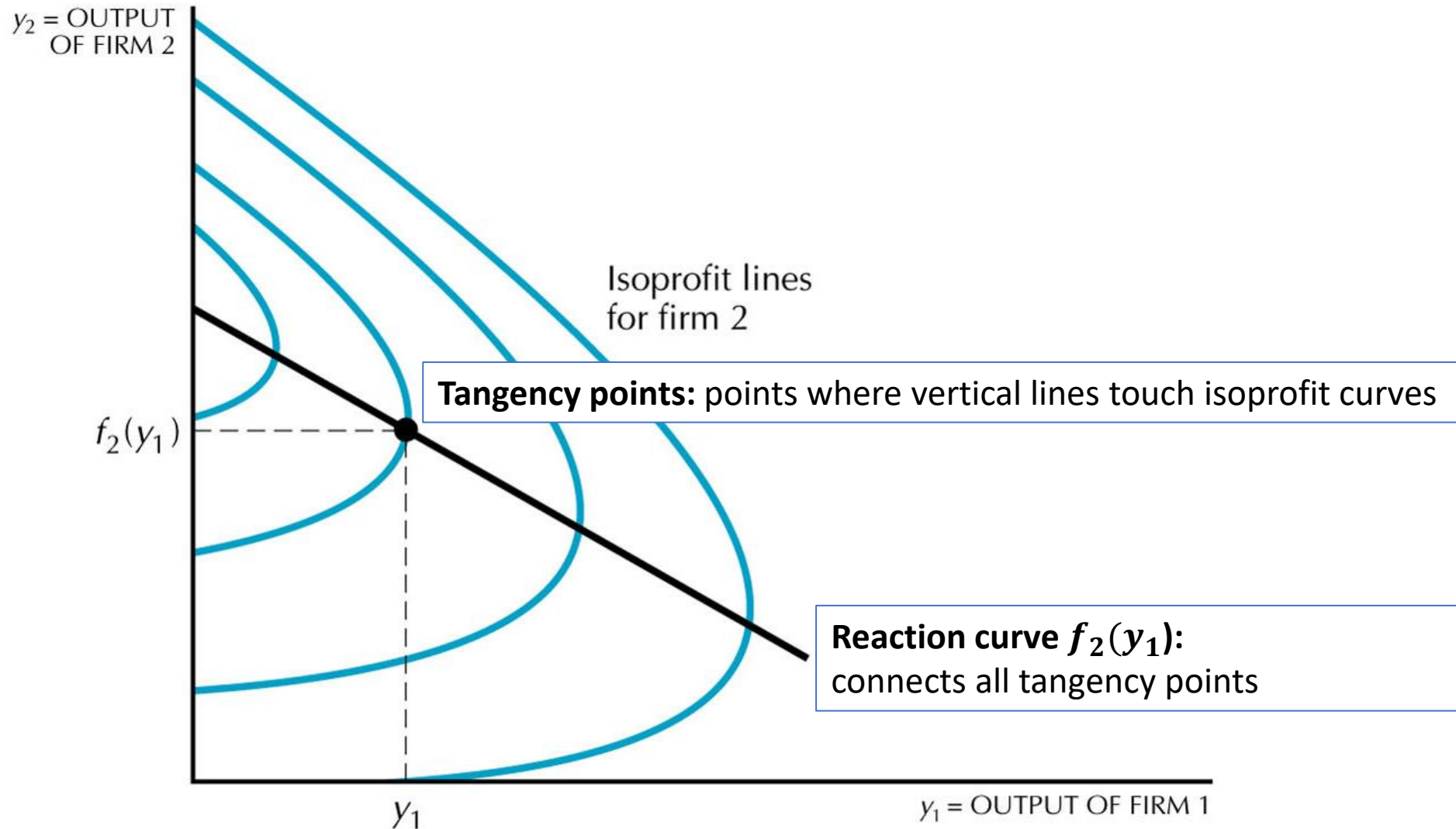


Sequential games usually solved from the end (backward induction)

Follower

- maximises $p(y_1 + y_2)y_2 - c_2(y_2)$ taking y_1 as a constant
- sets y_2 so that $MR_2(y_2) = p(y_1 + y_2) + \frac{\partial p(y_1 + y_2)}{\partial y_2} y_2 = MC_2(y_2)$

Sequential quantity setting (Stackelberg)



Sequential quantity setting (Stackelberg)

Follower with linear demand and constant marginal cost

Assumptions

- (Inverse) demand: $p(y_1 + y_2) = a - b(y_1 + y_2)$
- $MC_1 = MC_2 = c$

Solve

- $\pi_2(y_1, y_2) = [a - b(y_1 + y_2)]y_2 - cy_2 = ay_2 - by_1y_2 - by_2^2 - cy_2$
- Isoprofit lines: $ay_2 - by_1y_2 - by_2^2 - cy_2 = \overline{\pi_2}$
- Reaction curve: $\frac{\partial \pi_2}{\partial y_2} = a - by_1 - 2by_2 - c = 0 \Rightarrow y_2^*(y_1) = \frac{a - by_1 - c}{2b}$

Sequential quantity setting (Stackelberg)

Leader

- maximises $p(y_1 + y_2)y_1 - c_1(y_1) = p(y_1 + y_2^*(y_1))y_1 - c_1(y_1)$

With linear demand and constant marginal cost

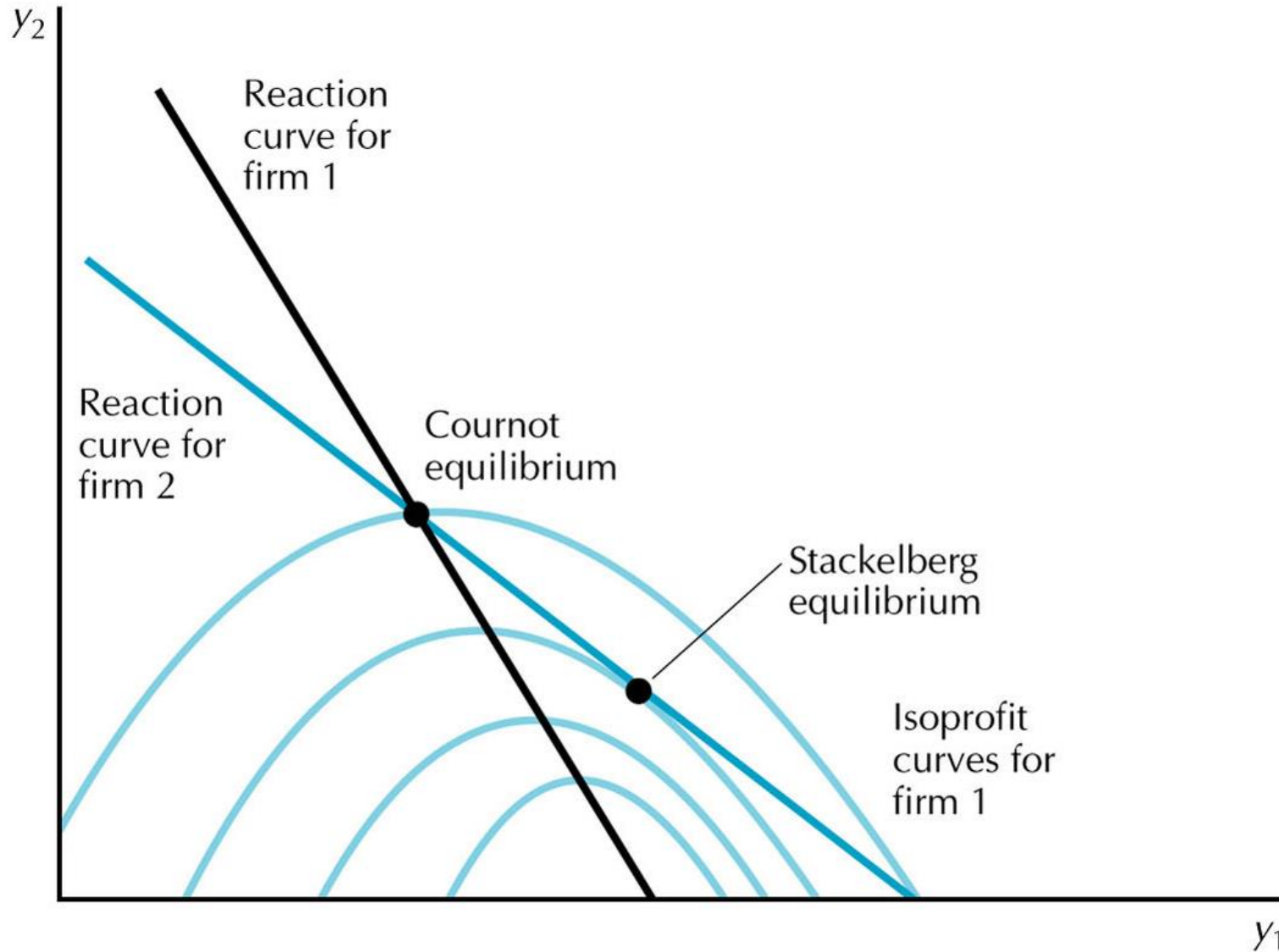
- $\pi_1(y_1, y_2) = [a - b(y_1 + y_2)]y_1 - cy_1 = ay_1 - by_1y_2 - by_1^2 - cy_1 = ay_1 - by_1 \frac{a - by_1 - c}{2b} - by_1^2 - cy_1 = \frac{ay_1 - by_1^2 - cy_1}{2}$
- $\frac{\partial \pi_1}{\partial y_1} = \frac{a-c}{2} - by_1 = 0 \Rightarrow y_1^* = \frac{a-c}{2b}$

Sequential quantity setting (Stackelberg)

Overview

- $y_1^* = \frac{a-c}{2b}$
- $y_2^* = \frac{a-c}{4b}$
- $Y^* = \frac{3(a-c)}{4b}$
- $p^* = \frac{a+3c}{4}$
- $\pi_1^* = \frac{(a-c)^2}{8b}$
- $\pi_2^* = \frac{(a-c)^2}{16b}$
- $\Pi^* = \frac{3(a-c)^2}{16b}$

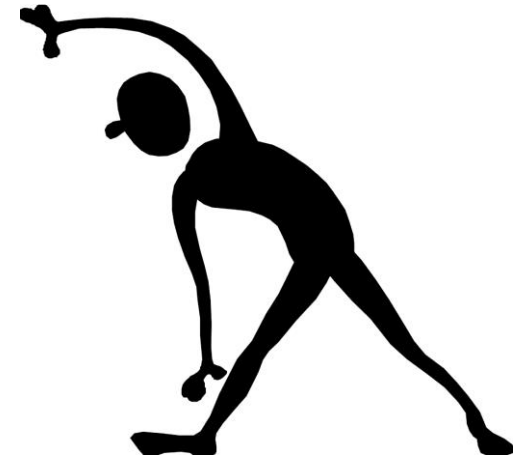
Sequential quantity setting (Stackelberg)



Sequential quantity setting (Stackelberg)

Exercise

- Market demand $D(p) = 12 - p$
- Cost functions
 - Leader: $c_1(y_1) = 2y_1 + 12$
 - Follower $c_2(y_2) = 4y_2 + 1$
- Questions:
 1. Derive the inverse demand and the profit and reaction functions of the follower.
 2. Derive the profit function of the leader and the Stackelberg equilibrium.
 3. Calculate the total output, price and profit of both producers.



Sequential quantity setting with more followers

- n firms (1 leader + (n-1) followers)

Followers (i=2,...,n)

- $\pi_i(y_1, \dots, y_n) = [a - b(y_1 + \dots + y_n)]y_i - cy_i = ay_i - by_i \sum_{j \neq i} y_j - by_i^2 - cy_i$
- $\frac{\partial \pi_i}{\partial y_i} = a - b \sum_{j \neq i} y_j - 2by_i - c = 0$
- Symmetricity of y_2 to $y_n \Rightarrow y_i = \frac{a-c}{nb} - \frac{y_1}{n}$

Leader

- $\pi_1(y_1, \dots, y_n) = \left[a - b \left(y_1 + (n-1) \left(\frac{a-c}{nb} - \frac{y_1}{n} \right) \right) \right] y_1 - cy_1 = \frac{ay_1 - by_1^2 - cy_1}{n}$
- $\frac{\partial \pi_1}{\partial y_1} = \frac{a-c}{n} - \frac{2by_1}{n} = 0 \Rightarrow y_1^* = \frac{a-c}{2b}$

Sequential quantity setting with more followers

Overview

- $y_1^* = \frac{a-c}{2b}$ (independent of n)
 - $y_2^* = \frac{a-c}{4bn}$ (decreasing in n)
 - $Y^* = \frac{(2n-1)(a-c)}{(2n)b}$ (increasing in n)
 - $p^* = \frac{a+(2n-1)c}{2n}$ (decreasing in n)
-
- $n=1 \rightarrow$ monopoly
 - $n \rightarrow \infty \rightarrow$ perfect competition

Simultaneous quantity setting (Cournot)

- 2 (or more) firms deciding quantity simultaneously

Examples

A



B



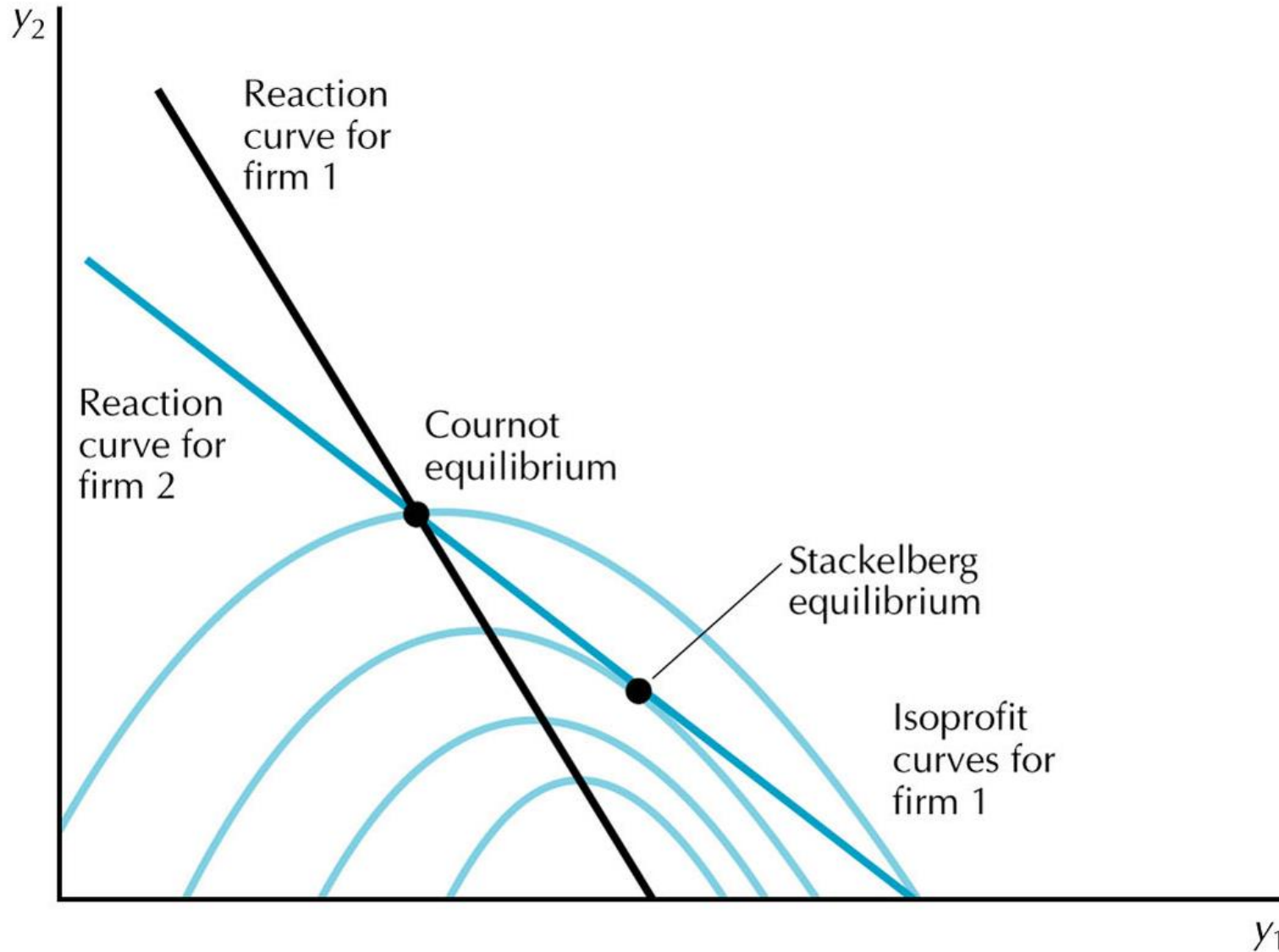
C



Simultaneous quantity setting (Cournot)

- firm 1 maximises $p(y_1 + y_2)y_1 - c_1(y_1)$ given expected y_2
- firm 1's reaction curve: what is the optimum y_1^* given expected y_2 ?
- Cournot equilibrium:
 - Pair of outputs such that each firm maximizes profit given its beliefs about the choice of the other firm.
 - The beliefs are confirmed.
 - Neither firm finds it profitable to unilaterally change its choice.
 - Point where the reaction curves cross.

Simultaneous quantity setting (Cournot)

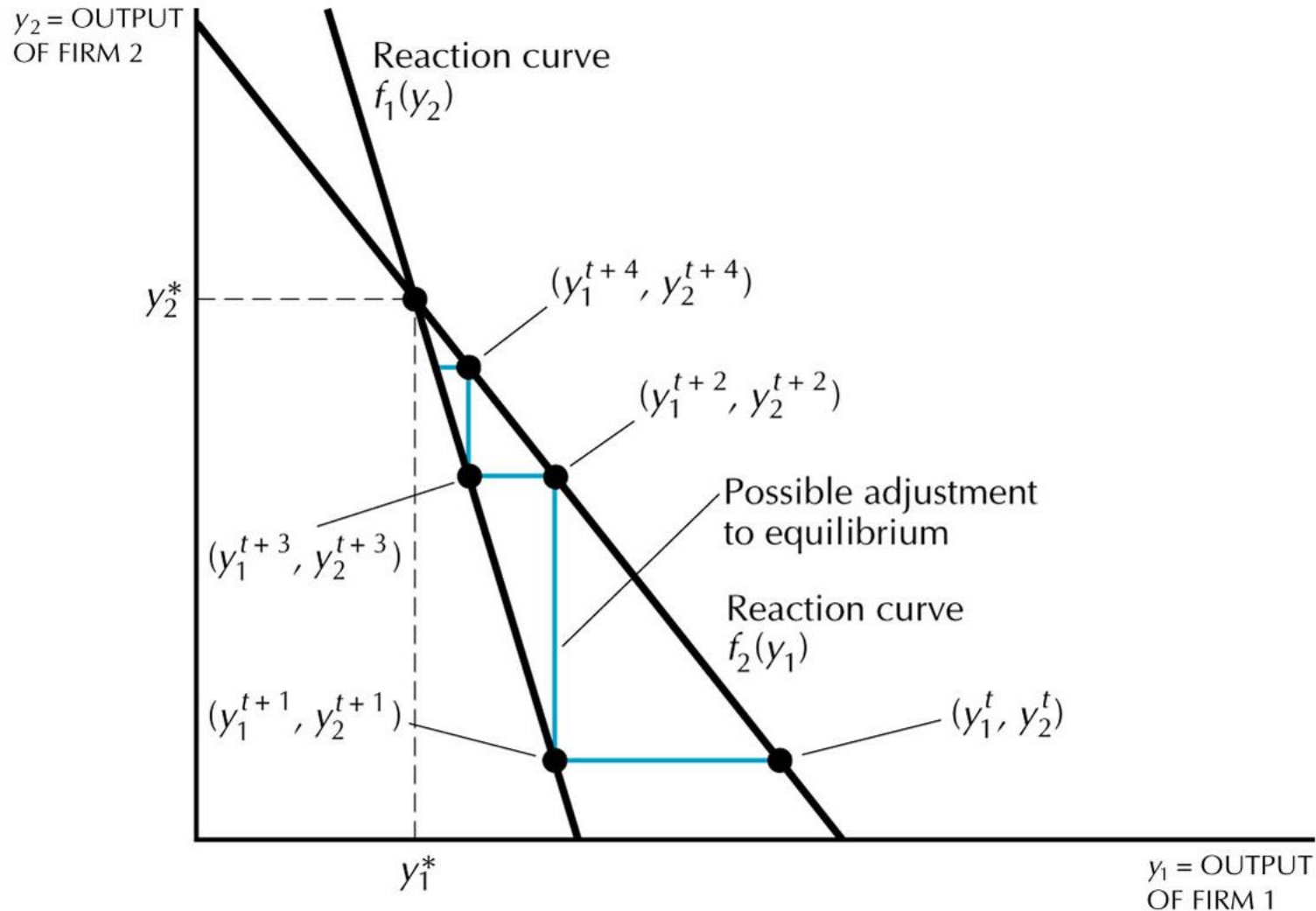


Simultaneous quantity setting (Cournot)

Case with linear demand and constant marginal cost

- Reaction curves $y_1^*(y_2) = \frac{a-by_2-c}{2b}$ and $y_2^*(y_1) = \frac{a-by_1-c}{2b}$
- Symmetric equilibrium:
 - $y_1^* = y_2^* = \frac{a-c}{3b}$
 - $Y^* = \frac{2(a-c)}{3b}$
 - $p^* = \frac{a+2c}{3}$
 - $\pi_1^* = \pi_2^* = \frac{(a-c)^2}{18b}$
 - $\Pi^* = \frac{(a-c)^2}{9b}$

Simultaneous quantity setting (Cournot)



Simultaneous quantity setting with more firms

- $\pi_i(y_1, \dots, y_n) = [a - b(y_1 + \dots + y_n)]y_i - cy_i = ay_i - by_i \sum_{j \neq i} y_j - by_i^2 - cy_i$
- $\frac{\partial \pi_i}{\partial y_i} = a - b \sum_{j \neq i} y_j - 2by_i - c = 0$
- Solution
 - $y_i^* = \frac{a-c}{(n+1)b}$ (decreasing in n)
 - $Y^* = \frac{n(a-c)}{(n+1)b}$ (increasing in n)
 - $p^* = \frac{a+nc}{n+1}$ (decreasing in n)
 - $\pi_i^* = \frac{(a-c)^2}{n(n+1)^2 b}$ (decreasing in n , very fast)
 - $\Pi^* = \frac{(a-c)^2}{(n+1)^2 b}$ (decreasing in n , fast)

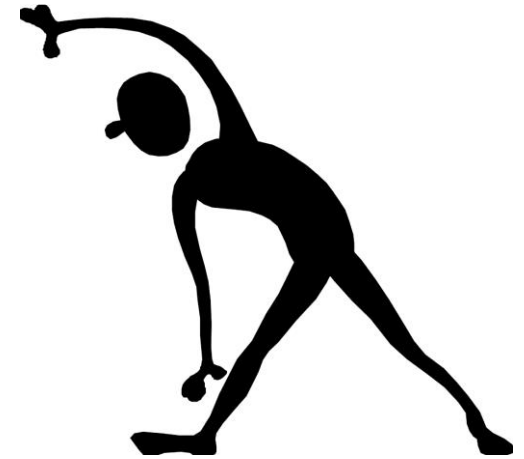
$n=1 \rightarrow$ monopoly

$n \rightarrow \infty \rightarrow$ perfect competition

Simultaneous quantity setting (Cournot)

Exercise

- Market demand $D(p) = 12 - p$
- Cost functions
 - Firm 1: $c_1(y_1) = 2y_1 + 12$
 - Firm 2: $c_2(y_2) = 4y_2 + 1$
- Questions:
 1. Derive profit functions of both firms.
 2. Derive the Cournot equilibrium.
 3. Calculate total output, price and profit of both producers.
 4. Compare to the total output and price in the Stackelberg equilibrium



Simultaneous price setting (Bertrand)

- 2 (or more) firms deciding price simultaneously

Examples

A



B



Simultaneous price setting (Bertrand)

- firm 1 maximises $y_1(p_1, p_2)p_1 - c_1(y_1(p_1, p_2))$ given expected p_2
- Bertrand equilibrium:
 - Pair of prices such that each firm maximizes profit given its beliefs about the choice of the other firm.
 - The beliefs are confirmed.
 - Neither firm finds it profitable to unilaterally change its choice.
- Finding the equilibrium
 - $p_i \geq MC$
 - If firm 1 sets $p_1 > MC$, firm 2 sets $MC < p_2 < p_1$ and steals all customers.
 - Firm 1 responds by undercutting firm 2 etc....
 - ➔ the only equilibrium is $p_1 = p_2 = MC$ ➔ the competitive equilibrium

(or, more accurately, $p=MC$ for all firms selling a strictly positive output)

What is the Bertrand equilibrium?

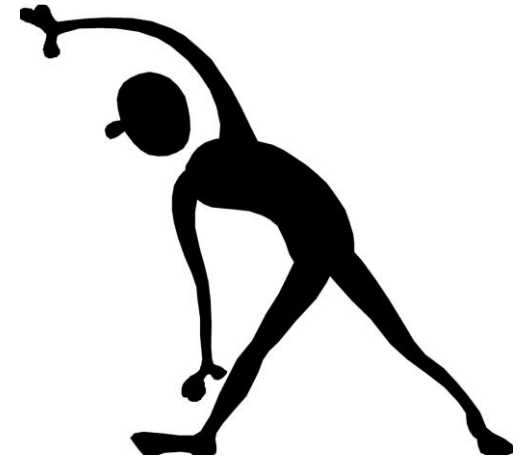
Simultaneous price setting (Bertrand)

- **Bertrand equilibrium with linear demand and homogeneous (constant) marginal costs**
 - $p_1^* = p_2^* = c$
 - $Y^* = \frac{a-c}{b}$
 - $\Pi^* = 0$
- **Bertrand equilibrium with heterogeneous constant marginal costs?**
 - The firm with the lowest MC get the entire market
 - Price equal to the 2nd lowest MC
- **Bertrand equilibrium with increasing marginal costs?**
 - Assumption: If the same MC, output divided equally
 - Output as in perfect competition

Simultaneous price setting (Bertrand)

Exercise

- Market demand $D(p) = 55 - \frac{p}{2}$
- Cost functions
 - Firm 1: $c_1(y_1) = y_1^2 + 293$
 - Firm 2: $c_2(y_2) = 2y_2^2 + 84$
- Questions:
 1. Derive the inverse demand function and marginal cost of both firms.
 2. Calculate each firm's output and price in the Bertrand equilibrium.
 3. Will both firms want to operate (in the longer run)?



Co-operative setting (collusion)

- **Collusion** = non-competitive agreement between competitors
- **Cartel** = a group of colluding firms

Examples

A



B



E-book price fixing

- **Background:**

- E-book market hit by Amazon
- Publishers threatened by cheap e-books
- Apple releasing the iPad
- Secret meetings by „Big 6“ CEOs
- Publishers approached by Apple

- **The deal:**

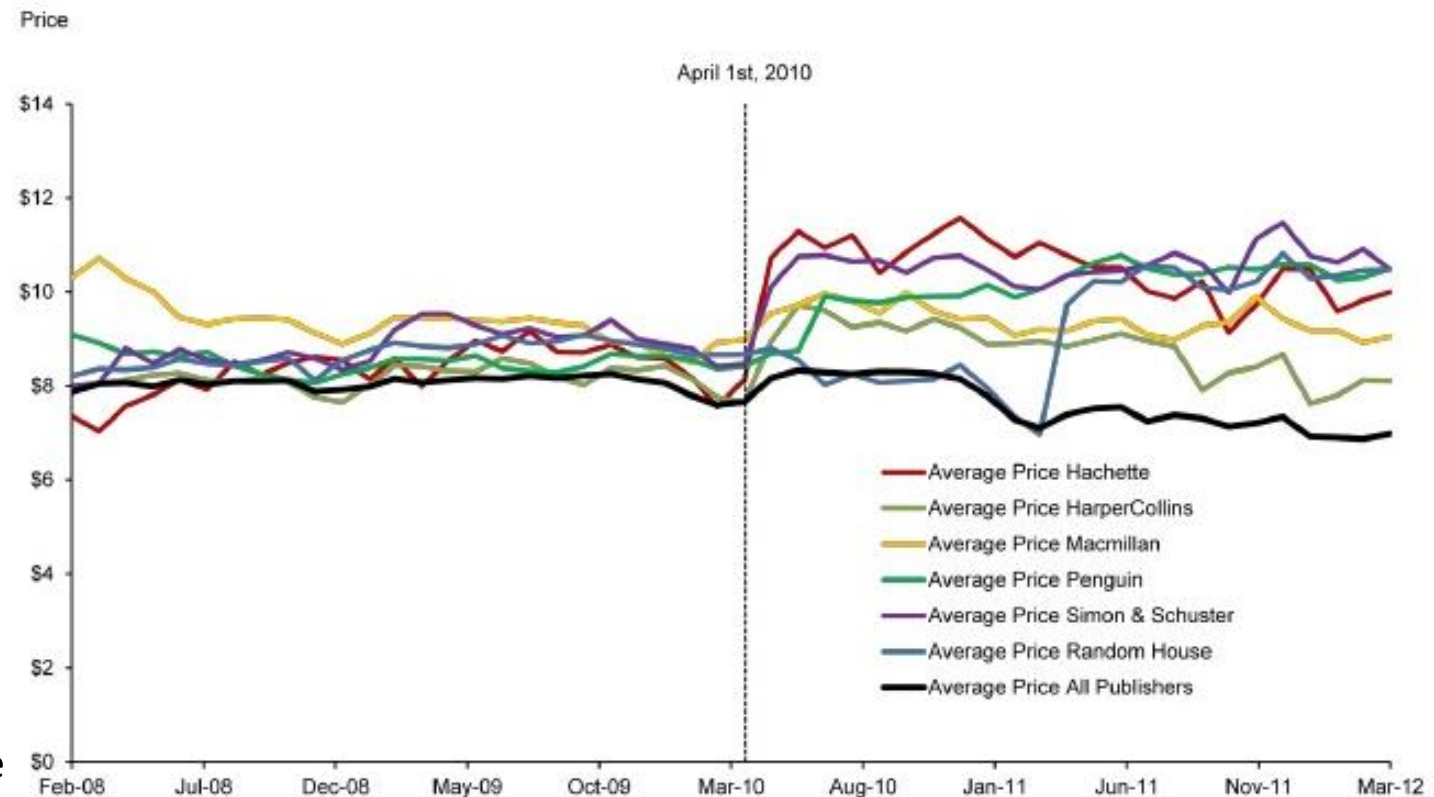
- In action 10th April 2010
- Sell e-books via iBookstore under an agency model (30% commission for Apple)
- Price tiers set up by Apple
- Increase prices of most books to 9.99-14.99
- Amazon also had to switch to the agency model (publishers co-ordinated in negotiations with Amazon)

- **Ruling:**

- The „Big 6“ US publishers + Apple conspired to raise the prices of e-books
- Apple to pay a fine of USD 450 millions

CONTAINS MATERIAL DESIGNATED AS HIGHLY CONFIDENTIAL PER PROTECTIVE ORDER

Average eBook Retail Prices for Sony, Amazon, Barnes & Noble, Apple, Google, Books-A-Million, and Kobo
February 2008 – March 2012



Source: Sony Transactions Data, Amazon Transactions Data, Barnes & Noble Transactions Data, Apple Transactions Data, Google Transactions Data, Books-A-Million Transactions Data, Kobo Transactions Data

Co-operative setting (collusion)

Analysis

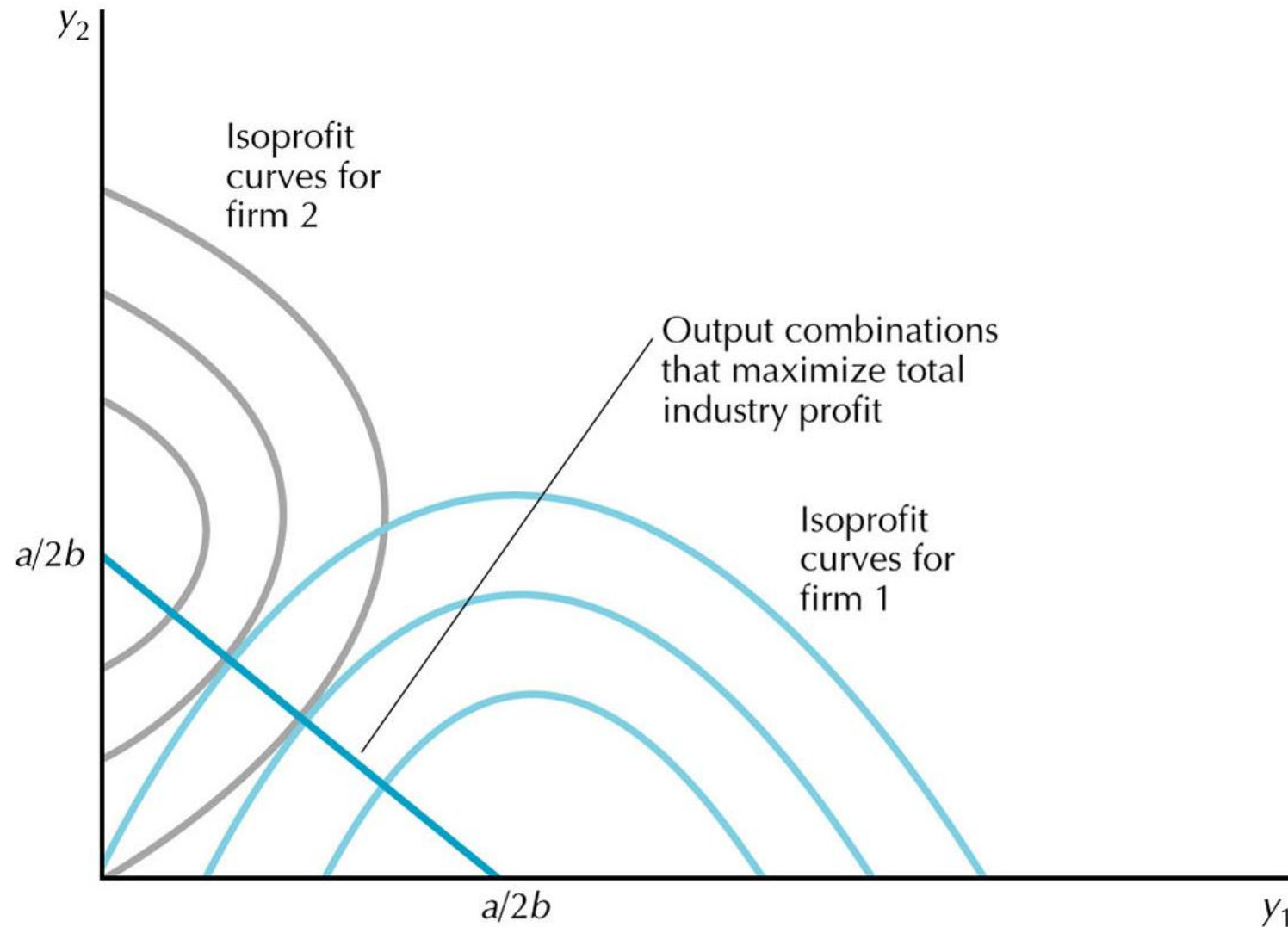
- producers jointly behave like a monopolist
- maximise $p(y_1 + y_2)(y_1 + y_2) - c_1(y_1) - c_2(y_2)$
- $\frac{\partial(\pi_1 + \pi_2)}{\partial y_1} = p(y_1 + y_2) + \frac{\partial p(y_1 + y_2)}{\partial y_1} (y_1 + y_2) - MC_1(y_1) = 0$
- In the optimum, marginal cost is the same: $MC_1 = MC_2 = p(Y) + \frac{\partial p(Y)}{\partial Y}$
 - The firm with higher MC (at given output) has lower profit and may not be willing to participate in the cartel
 - Compensation may solve this, but is difficult given that the cartel is illegal

Co-operative setting (collusion)

With linear demand and zero marginal costs

- $\Pi(y_1 + y_2) = [a - b(y_1 + y_2)](y_1 + y_2) = a(y_1 + y_2) - b(y_1 + y_2)^2$
- $\frac{\partial \Pi}{\partial Y} = a - 2b(y_1^* + y_2^*) = 0$
- Solution
 - $Y^* = \frac{a}{2b}$
 - $p^* = \frac{a}{2}$
 - $\Pi^* = \frac{a^2}{4b}$

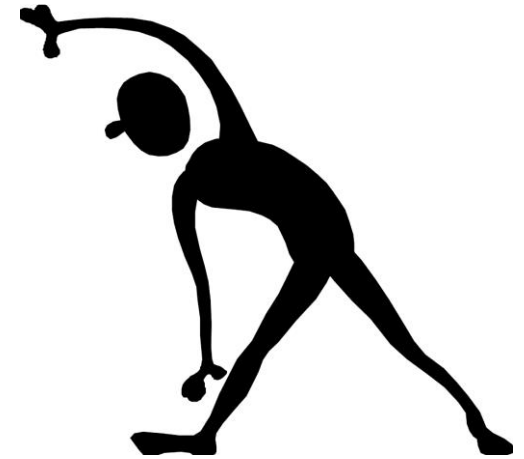
Co-operative setting (collusion)



Co-operative setting (collusion)

Exercise

- Market demand $D(p) = 230 - p$
- Cost functions
 - Firm 1: $c_1(y_1) = y_1^2 + 100$
 - Firm 2: $c_2(y_2) = 2y_2^2 + 100$
- Questions:
 1. Derive the Cournot equilibrium quantities and price.
 2. Derive the equilibrium quantities and price in the case of a collusion.
 3. Compare the total output and price in the two situations.
 4. Will both firms want to participate in the cartel?



Co-operative setting (collusion)

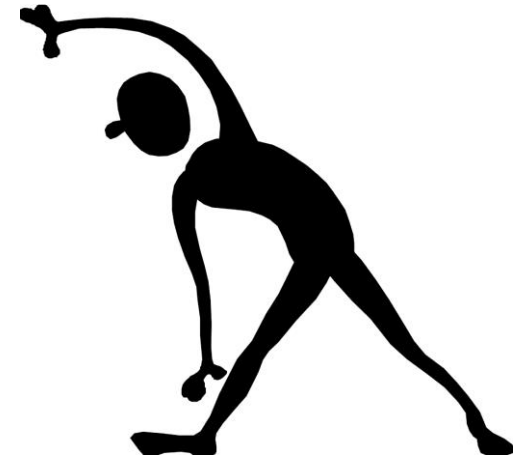
Temptation to cheat

- Marginal profit when firm 1 increases its output above the agreed quantity:
 - $\frac{\partial \pi_1}{\partial y_1} = p(y_1^* + y_2^*) + \frac{\partial p(y_1 + y_2)}{\partial y_1} y_1^* - MC_1(y_1^*)$
 - But when joint profit maximised: $\frac{\partial \Pi}{\partial y_1} = p(y_1^* + y_2^*) + \frac{\partial p(y_1 + y_2)}{\partial y_1} (y_1^* + y_2^*) - MC_1(y_1^*) = 0$
- $\frac{\partial \pi_1}{\partial y_1} = -\frac{\partial p(y_1 + y_2)}{\partial y_1} y_2^* > 0$ Firm 1 can increase its profits by producing more than agreed!

Co-operative setting (collusion)

Exercise

- Market demand $p = 90 - 2Y$
- Cost functions
 - Firm 1: $c_1(y_1) = y_1^2 + 20$
 - Firm 2: $c_2(y_2) = y_2^2 + 20$
- Questions:
 1. What will be the profit of both firms if they maximise joint profit and split it evenly?
 2. What will be their profits if firm 1 cheats and firm 2 does not?
 3. What will be their profits if they both cheat?



Comparison of oligopoly settings

(with linear demand and homogenous products)

$$p^{Bertrand} = c < p^{Stackelberg} = \frac{a + 3c}{4} < p^{Cournot} = \frac{a + 2c}{2} < p^{Monopoly} = \frac{a + c}{2}$$

$$Y^{Bertrand} = \frac{a-c}{b} > Y^{Stackelberg} = \frac{3(a-c)}{4b} > Y^{Cournot} = \frac{2(a-c)}{3b} > Y^{Monopoly} = \frac{a-c}{2b}$$

$$\Pi^{Bertrand} = 0 < \Pi^{Cournot} = \frac{(a-c)^2}{9b} < \Pi^{Stackelberg} = \frac{3(a-c)^2}{16b} < \Pi^{Monopoly} = \frac{(a-c)^2}{4b}$$



Note that Stackelberg implies lower price and greater total output than Cournot, but also greater total profits

Differentiated (heterogeneous) products

- In many real-world markets they differentiate their products (substitutes), in terms of quality, style, branding, guaranties, advertising, etc.

- **Horizontal** differentiation: e.g.



- **Vertical** differentiation: e.g.



VS.



- As a result they can keep part of a market even if they charge different prices.
- We discuss 3 settings with (horizontal) product differentiation
 - Price setting with heterogeneous products (Bertrand)
 - Spatial model of horizontal differentiation
 - Monopolistic competition

Price setting with differ. products (Bertrand)

Motivation

- Price setting realistic in many markets
- But Bertrand model with homogenous products gives rather extreme predictions
- Bertrand model more realistic with differentiated products

Setup

- Imperfectly substitutable products with symmetric demands
 - $q_1 = a - bp_1 + hp_2$, and
 - $q_2 = a - bp_2 + hp_1$
- where $b > h$

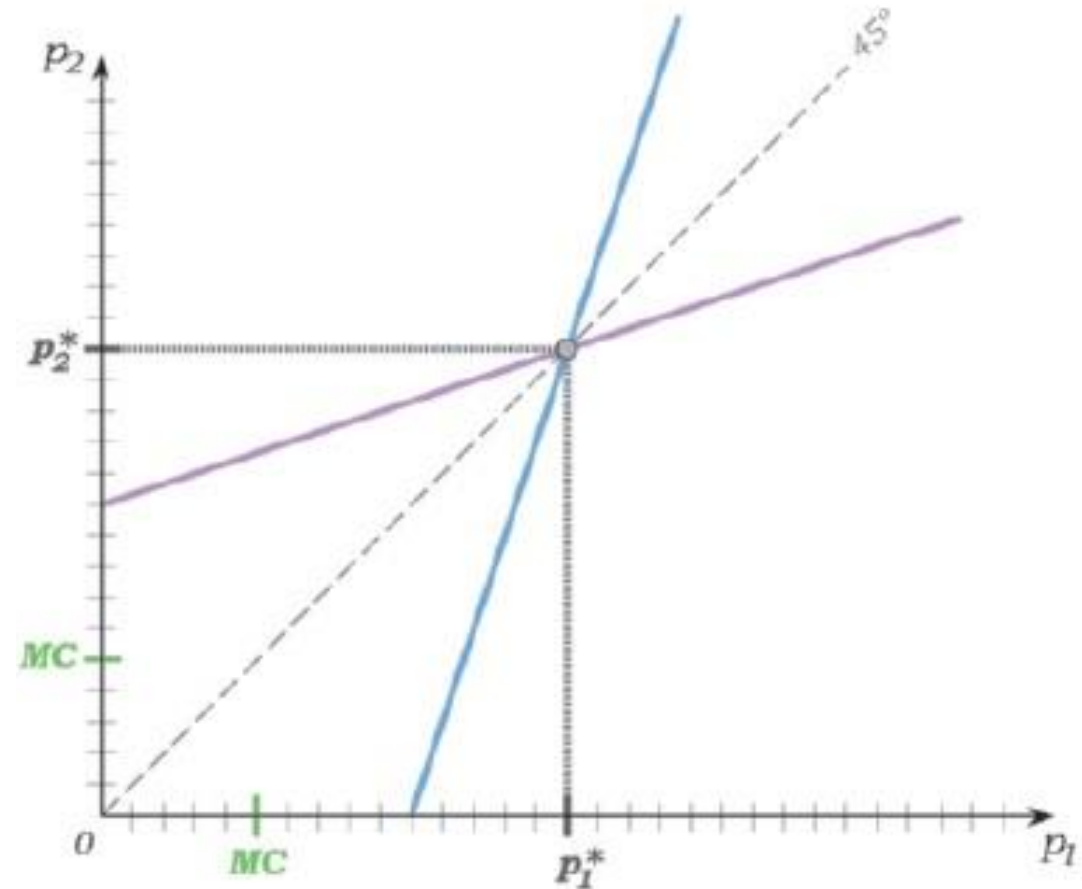
Price setting with differ. products (Bertrand)

Analysis

- Firm 1 maximises $\pi_1(p_1, p_2) = (p_1 - c)(a - bp_1 + hp_2)$
- FOC: $\frac{\partial \pi_1}{\partial p_1} = a - 2bp_1 + hp_2 + cb = 0$
- Best-response functions:
 - $p_1^*(p_2) = \frac{a+cb+hp_2}{2b}$
 - $p_2^*(p_1) = \frac{a+cb+hp_1}{2b}$

Solution

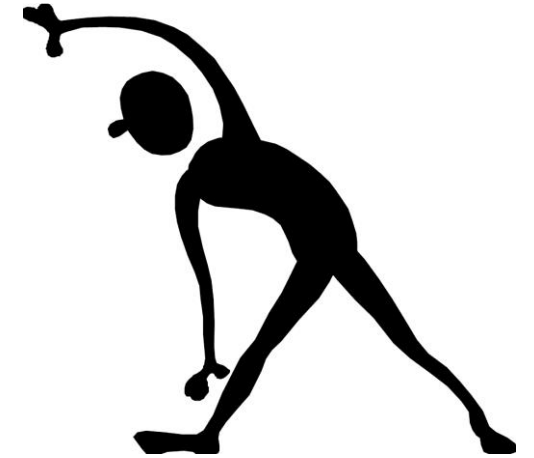
- $p_1^* = p_2^* = \frac{a+cb}{2b-h}$
- $q_1^* = q_2^* = \frac{b(a-c(b-h))}{2b-h}$
- Assuming $a > c(b-h)$ (so $q_1^* = q_2^* > 0$) $\rightarrow p > c$



Price setting with differ. products (Bertrand)

Exercise

- Demand functions
 - Firm 1: $q_1(p_1, p_2) = 90 - 2p_1 + p_2$
 - Firm 2: $q_2(p_1, p_2) = 90 - 2p_2 + p_1$
- $MC=0$
- Questions:
 1. Derive the best-response functions of both firms
 2. Derive the equilibrium quantities produced and prices charged
 3. Derive the profit of both firms



Spatial model of horiz. differentiation (Hotelling)

Setup

- Product **differentiation endogeneously determined** by firms
- Differentiation modeled as **product location**
- **Consumers** also characterised by the location of each consumer
 - Consume one unit of the good
 - Incur transportation costs which depend on the distance to firm
- Literal interpretation (**physical** location) or metaphorical interpretation (location in the **product space**)
- Two-stage game:
 1. Firms choose location
 2. Firms choose price where location is chosen first and price afterwards.
- **Intuition:** firms choose maximum product differentiation to minimise price competition in 2nd stage

Spatial model of horiz. differentiation (Hotelling)

1 firm

- Locates in middle of consumer preferences
- Socially optimal



Spatial model of horiz. differentiation (Hotelling)

2 firms

- Have incentive to move towards the other firm (they gain extra consumer but do not lose any)
- In equilibrium, both firms in the middle
- Inefficient, as a socially optimal location would be in $1/4$ and $3/4$ of the line



Spatial model of horiz. differentiation (Hotelling)

3 firms

- The outside firms have an incentive to get towards the middle one
- But once they get very close, the middle one has an incentive to skip towards the edge
- No (pure strategy) equilibrium



> 3 firms

- (Pure strategy) equilibria exist

Spatial model of horiz. differentiation (Hotelling)

Examples

- Ice-cream vendors on a beach



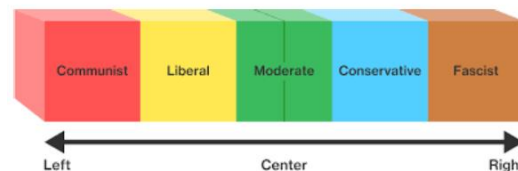
- Petrol stations along a highway



- Posh vs. cheap restaurants



- Political parties



Monopolistic competition

Setup

- Products are substitutes but differ (e.g. by features, brand etc.)
- Elements of both competition and monopoly
- **Like a monopoly**
 - Each firm faces a **downward sloping demand curve**
 - The stronger the product differentiation, the less elastic the demand curve
- **Like perfect competition**
 - The demand curve depends on **output and price decisions of other firms**
 - **Each firm too small** to consider its effects on other firms
 - **Free entry**
 - Demand curve **shifts inwards** as more firms enter
 - Demand curve **becomes more elastic** as products become more similar across firms

Monopolistic competition

Examples

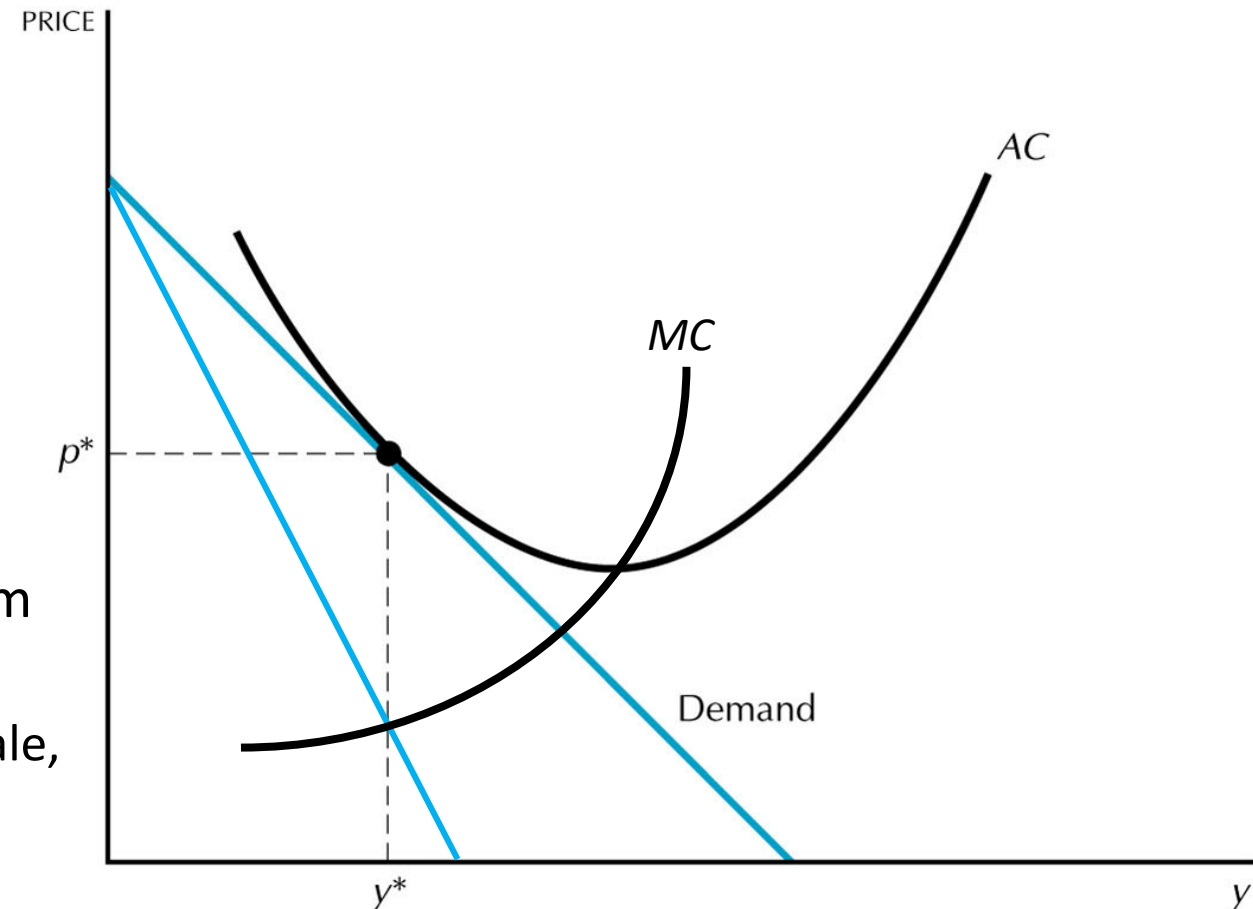
- Many manufacturing industries
 - E.g. toys, tableware etc. etc.
- Many services:
 - E.g. accountants, therapists...



Monopolistic competition

Analysis

- Equilibrium conditions
 1. Each firm selling at a price and output combination on its demand curve
 2. Firms maximize profit ($MR=MC$)
 3. Free entry pushes profit to zero.
- D and AC curves are tangent.
- Pareto inefficient situation as $p > MC$
- Firms operate to the left of the AC minimum
 - „Excess capacity“
 - Fewer firms would mean more efficient scale, but lower variety.



Group discussion

Please get in groups of 3.

1. Are firms in the following oligopoly markets better described by quantity or price competition?
 - travel insurance; oil; new apartments; mobile plans
2. What are the key differences between assumptions and implications of the Stackelberg and Cournot oligopoly models?
3. Why are drug cartels called cartels?
4. Give examples of horizontal and vertical product differentiation in each of the following industries:
 - clothes; mobile phones; master's study programmes
5. What are the key differences between monopolistic competition vs. oligopoly?